KWAME NKRUMAH UNIVERSITY OF SCIENCE AND TECHNOLOGY, KUMASI

COLLEGE OF ENGINEERING

ME 362 VIBRATIONS I

Assignment 1

Due Date: Tuesday, 13th June, 2024 Time: Before Class

1. Figure 1 shows a uniform rigid bar of mass 10 kg that is pivoted at point O and connected by springs of stiffnesses $k_1=10$ kN/m and $k_2=20$ kN/m and considering a small angular displacement of the rigid bar about the point O, determine the range of values of l over which the bar will be stable.

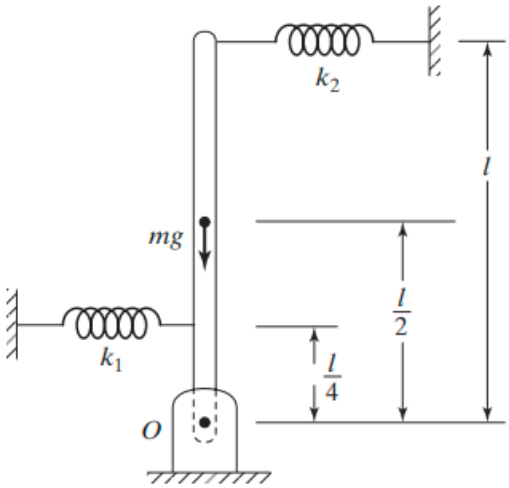


Figure 1

2. A gear wheel has a diameter of 5 m and a mass of 9t. It is suspended from knife-edge so that it is free to swing in a vertical plane at right angles to the axis of the gear. The point of suspension is 100 mm inside the pitch circle. The period of one complete oscillation is 4.29 s.
   Determine: (a) the moment of inertia of the gear about its axis of symmetry, stating the units carefully and justifying any compound pendulum equation used, and (b) the tangential force that must be applied at the pitch radius of this gear to give it an angular acceleration of 0.9 rad/s^2.

3. The motion of a 10 kg pendulum is controlled by a spring and dashpot, as shown in Fig. 2. The spring stiffness is 5 kN/m and the dash-pot piston has a mass of 0.75 kg. The bell-crank lever has a mass of 2.5 kg and a radius of gyration of 50 mm about its centre of gravity which is at the fulcrum. When the mass is displaced from its equilibrium position and released, the amplitude of the resulting vibration is reduced by one-half over three vibrations. Determine the damping coefficient at the dash-pot, and the value required to give critical damping. Also find the response of the system if the initial angular velocity and accelerations are $\pi/50$ rad/s and $\pi/20$ rad/s^2 respectively.

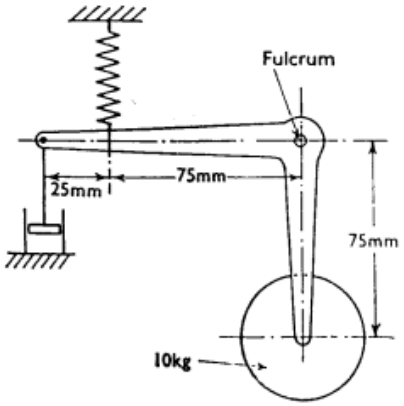


Figure 2

4. A fan is mounted on a spring and viscous damper in parallel so that only linear motion in the vertical direction occurs. Briefly derive an expression for the force transmitted to the ground through the spring and damper if the fan generates an harmonic disturbing force in the vertical direction. Given that the fan has a mass of 40 kg and a rotating unbalance of 0.01 kg m, determine the spring stiffness required for 50% force transmission. Take the damping ratio of the system to be 0.2 and the fan speed as 1200 rev/min. When running under these conditions what effect on the transmission would removal of the damping element have?

5. Figure 3 shows a simple model of a motor vehicle that can vibrate in the vertical direction while traveling over a rough road. The vehicle has a mass of 1200 kg. The suspension system has a spring constant of 400 kN/m and a damping ratio of 0.5. If the vehicle speed is v km/hr, Find the Fourier series expansion of y(t) using Furies Series and determine the steady-state displacement amplitude of the vehicle.

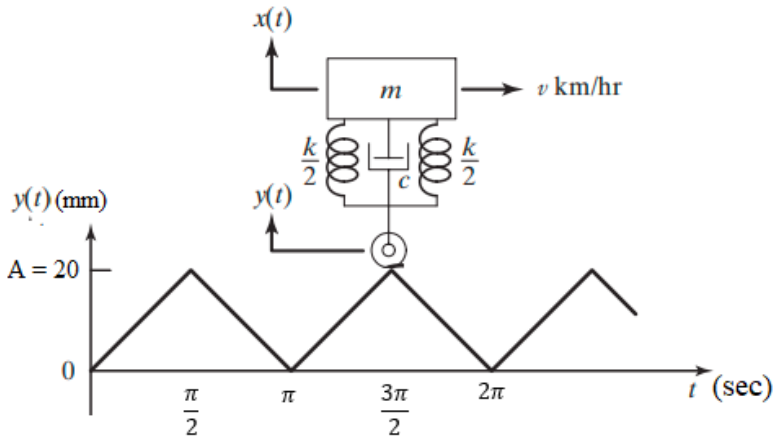


Figure 3

6. A steel shaft, 60 mm diameter, with its ends freely supported in bearings 0.9 m apart, carries two wheels, each of mass 50 kg and each placed 0.3 m from one of the bearings. Calculate the whirling speed of the shaft, assuming the shaft effect to be equivalent to 17/35ths of its mass concentrated at mid-span. Take ρ=7.8 Mg/m^3 and E=200 GPa for steel.